

EC4219: Software Engineering

Lecture 16 — Wrap-up

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Software and Software Engineering¹

- **Software:** is a collection of computer programs and associated documentation.
 - ▶ E.g., specification, instruction manual
 - ▶ In this course, we will interchangeably use the terms between program and software.
- **Software Engineering:** is an **engineering discipline** that is concerned with all aspects of **software production process**. The goal is to achieve cost-effective software development.

¹ “Software Engineering” by Sommerville

Software Production Process²

- A structured set of activities required to develop a software system.
- Examples include:
 - ▶ Specification: defining what the system should do.
 - ▶ Design and Implementation: defining the organization of the system and implementing the system.
 - ▶ Validation: checking that the system meets what the customer wants.
 - ▶ Evolution: changing the system in response to changing customer needs.
- In this course, **we focused on studying the validation** for ensuring software safety, correctness, and quality.
 - ▶ Why? The validation (SW testing and debugging) is the key. It comprises over 50% of the cost of software development.
 - ▶ Why should we invest so much in validation? Why is it so important? SW is written by humans, and humans make mistakes.

² “Software Engineering” by Sommerville

Topics Covered

- **Computational logic:** propositional logic, first-order logic, first-order theories
- **Formal methods:** Program verification, symbolic execution, type system

cf) Not covered: fuzzing, abstract interpretation, model checking, etc.

- Q. Did we learn something meaningful? Why not just use LLMs?

Closing

- Q. Did we learn something meaningful? Why not just use LLMs?
- Unlike formal methods, LLMs cannot formally prove/disprove SW security & safety.
 - ▶ Why? due to their statistical nature — (can only produce “likely” answers)
- We learned fundamental techniques in class – I hope you enjoyed it!